

Notes for ‘Inverse functions’

Important Ideas and Useful Facts:

- (i) **Vertical line test:** A curve in the xy -plane is the graph of some function f if and only if it satisfies the *vertical line test*, which means that as a vertical line scans from left to right, it only passes through the curve in at most one place.

In this case, a potential input on the x -axis can correspond to at most one potential output on the y -axis, which then becomes the unique value $f(x)$ of the function.

- (ii) **Horizontal line test and inverse functions:** If a horizontal line scanning from top to bottom only passes through the graph of a function $y = f(x)$ in at most one place, then we say that the graph satisfies the *horizontal line test* and that the function f is *one-one*. This means that each potential output on the y -axis can correspond to at most one input on the x -axis.

Suppose then that the graph of a function $y = f(x)$ satisfies the horizontal line test (and the vertical line test automatically, since f is a function). By reflecting the graph in the line $y = x$, we interchange the roles of x and y , and horizontal and vertical directions, creating a reflected image that satisfies the vertical line test. This reflected image then becomes the graph of a new function, called the *inverse* of f , denoted by f^{-1} . The rule for f^{-1} “undoes” or “reverses” the rule for f :

$$y = f(x) \quad \text{if and only if} \quad x = f^{-1}(y) .$$

A function whose graph does not satisfy the horizontal line test may have its domain restricted so that it does become one-one, illustrated below with the square function.

- (iii) **Interchanging domain and range:** As one passes from a function f to its inverse function f^{-1} , the roles of inputs and outputs are reversed, so naturally the roles of domain and range are interchanged, that is, the domain of f^{-1} equals the range of f , and the range of f^{-1} equals the domain of f .
- (iv) **Algebraic method for inverting functions:** To invert $y = f(x)$, one may rearrange the rule to express x in terms of y , to get $x = f^{-1}(y)$, and then finally revert to using x as a typical input.

For example, suppose that the function f is given by the rule

$$y = f(x) = \frac{\sqrt{x+2} - 1}{3} .$$

We rearrange this in several steps to get x expressed in terms of y :

$$\begin{aligned} y = \frac{\sqrt{x+2} - 1}{3} &\implies 3y = \sqrt{x+2} - 1 \implies 3y + 1 = \sqrt{x+2} \\ &\implies x + 2 = (3y + 1)^2 \implies x = (3y + 1)^2 - 2 . \end{aligned}$$

This shows that $f^{-1}(y) = x = (3y + 1)^2 - 2$, giving an explicit rule in terms of an input y . Reverting to using x as a typical input, we get the following rule for f^{-1} :

$$f^{-1}(x) = (3x + 1)^2 - 2 .$$

Examples and derivations:

1. Consider the square function $y = f(x) = x^2$, defined for all real x . Its domain is $\mathbb{R}$ and its range is the interval $[0, \infty)$. Its graph is the standard parabola with apex at the origin.

Except at the apex, any horizontal line passing through the graph intersects the parabola in two places, so the horizontal line test fails, so the function is not one-one.

To create an invertible function, we could, for example, restrict the domain of f to $[0, \infty)$, considering only the right half of the parabola. Then the horizontal line test would be satisfied, so that f becomes one-one and has an inverse f^{-1} . In this case the rule for the inverse function would be

$$f^{-1}(x) = \sqrt{x},$$

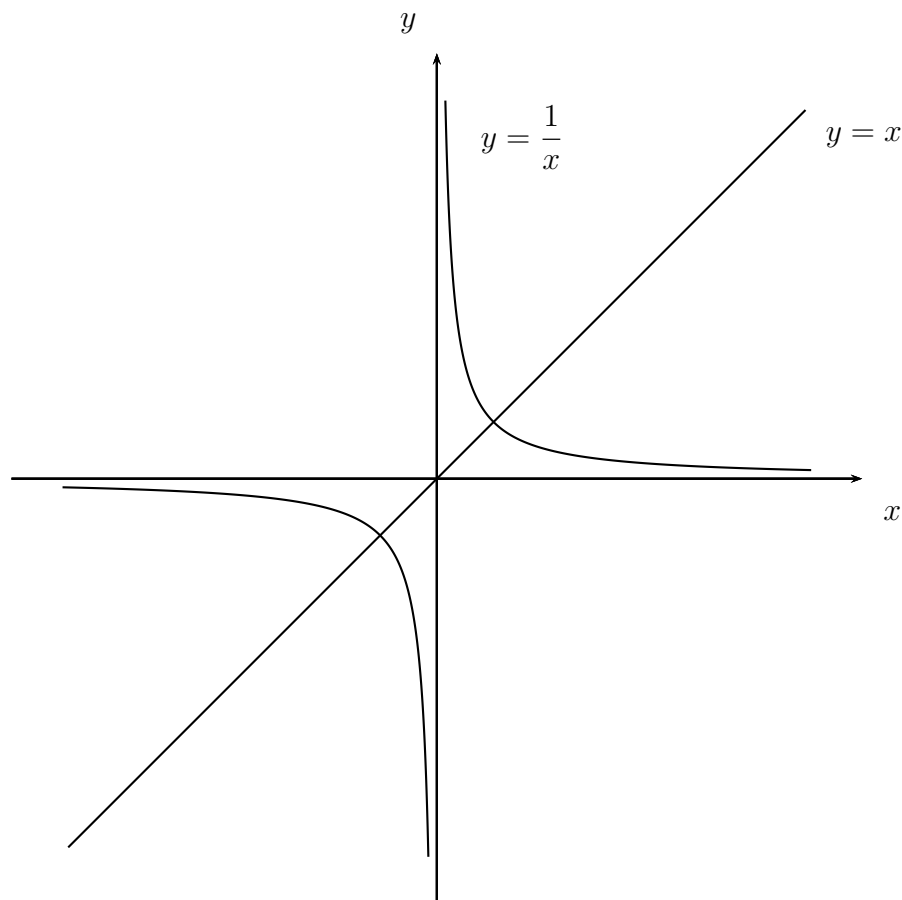
the positive square root function. The domain and range of both functions is the interval $[0, \infty)$.

By contrast, we could restrict the domain of f to $(-\infty, 0]$, considering the left half of the parabola. Again the horizontal line test would be satisfied. In this case the rule for the inverse function would be

$$f^{-1}(x) = -\sqrt{x},$$

the negative square root function. Now the domain of f and the range of f^{-1} are equal to the interval $(-\infty, 0]$, whilst the range of f and the domain of f^{-1} are equal to the interval $[0, \infty)$.

2. Consider the function h whose rule is $h(x) = \frac{1}{x}$ and whose graph is the hyperbola pictured below, together with the line $y = x$:



The domain and range of h are both equal to $\mathbb{R} \setminus \{0\}$, the real line with zero removed. It is clear that the graph satisfies both the horizontal and vertical line tests, so h is invertible. Moreover, the graph of h has perfect reflectional symmetry about the line $y = x$, so will coincide with itself upon reflection. Thus h and h^{-1} should coincide and have the same rules. In particular h should “undo” itself, and we can check that indeed composing the rule for h with itself, that is, forming $h \circ h$, should return an input $x \in \mathbb{R} \setminus \{0\}$ back to itself:

$$(h \circ h)(x) = h(1/x) = \frac{1}{1/x} = x ,$$

verifying indeed that $h \circ h$ is the identity function. Hence $h = h^{-1}$.

3. We explain briefly the reason why the notation f^{-1} is used to denote the inverse function corresponding to f .

For simplicity, to avoid any delicate issues related to domains of definition, we may assume f and f^{-1} both have the same domain D , which is some subset of $\mathbb{R}$.

(For example, in the previous example, the functions h and h^{-1} , which happened to coincide, both had domain $D = \mathbb{R} \setminus \{0\}$.)

The idea is that f and f^{-1} “undo” each other in the sense that following one by the other, applied to a given input, should unravel the effects of either function, getting back to the original input. Thus, for any input $x \in D$, we should have

$$(f^{-1} \circ f)(x) = f^{-1}(f(x)) = x \quad \text{and} \quad (f \circ f^{-1})(x) = f(f^{-1}(x)) = x .$$

Denote the identity function applied to the domain D by $\mathbf{1}$, with rule $\mathbf{1}(x) = x$ for all $x \in D$. Then these two equations are saying that

$$(f^{-1} \circ f)(x) = x = \mathbf{1}(x) \quad \text{and} \quad (f \circ f^{-1})(x) = x = \mathbf{1}(x)$$

for all $x \in D$. Thus, since the rules coincide exactly, we have produced the following two equations in the “arithmetic” of functions, employing now the circle operation $\circ$:

$$f \circ f^{-1} = \mathbf{1} = f^{-1} \circ f .$$

But these equations are entirely analogous to the following equations in ordinary arithmetic, employing usual multiplication $\times$, for any nonzero real number α , whose reciprocal $\frac{1}{\alpha}$ is denoted by α^{-1} :

$$\alpha \times \alpha^{-1} = 1 = \alpha^{-1} \times \alpha .$$

In this analogy, we think of the circle operation $\circ$ as corresponding to ordinary multiplication $\times$ and the identity function $\mathbf{1}$ as corresponding to the real number 1.

In courses in advanced abstract algebra, the arithmetic of invertible functions, working over a fixed domain D , is called a *group*, and the circle operation satisfies certain axioms, including the above equation involving f , f^{-1} and the distinguished element $\mathbf{1}$.